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Small Project of Selected Topic of Data Science

UNDERSTANDING ZERO GRADES: PATTERNS, ASSOCIATED FEATURES, AND ABSENCE–GRADE SENSITIVITY IN MATHEMATICS AND PORTUGUESE

Haqi Ardika Ramadani	5002231016
Reza Dwi Rahayu	5002231049
Layli Nurul Fatimah Zahra	5002231145

Group 6

Motivation

Why 0 Grades Matter?

- A grade of 0 may not be a real exam result.
- It may mean dropout, transfer, or withdrawal.
- The final grade has 53 zero records (9.6% in Math, 2.3% in Portuguese).

Why We Need to Analyze it?

- The dataset has no label for what a zero means.
- Treating zeros as ordinary scores can mislead the analysis.

Challenge

No label explains why a grade is 0 (dropout, transfer, absent from exam, or a genuine score).

Zeros sit on the same 0–20 scale as valid grades, so they cannot be told apart by value alone.

Zeros distort absence-grade correlations

Two overlapping subjects, no student ID

IDEA

classify zeros into Type A (G2 = G3 = 0), Type B (G3 = 0, G2 > 0), and a G1 = 0 anomaly.

compare G1, G2, absences, and other features across zero types and normal students.

identify important factors of academic achievement, with and without zeros

measure the absences–G3 correlation under three scenarios (all data, all G3 = 0 removed, only Type A removed) with bootstrap confidence intervals.

Research Goals

1

To **identify the pattern of zero grades in G1, G2, and G3** in both subjects (frequency, combinations across stages, and persistence), and to assess whether a zero behaves like an academic grade or like a marker of exiting the assessment.

2

To explore the **relationship between zero grades and student characteristics** (past failures, study time, learning support, and others) at comparable levels of initial grade, comparing Mathematics and Portuguese.

3

To assess the sensitivity of the correlation between student absences and final grades to the inclusion or exclusion of students with zero final grades.

DATA
UCI STUDENT PERFORMANCE DATASET

395
Mathematics Students

649
Portuguese Students

Zero Final Grades (G3 = 0)
9,6% Mathematics
2,3% Portuguese

DATASET OVERVIEW

Source
UCI Machine Learning Repository (Cortez & Silva, 2008)

Schools
Two Portuguese secondary schools (GP (Gabriel Pereira) and MS (Mousinho da Silveira))

Records
395 Mathematics | 649 Portuguese

Features
33 attributes per file, including demographic, social, school related features, and student grades (G1, G2, G3)

Data Origin
Grades and absences from school reports; other attributes from questionnaires

DATA

Zero-Grade Patterns and Classification

TYPE A

G2 = G3 =0

Examines whether zero grades persist across assessment periods. All 13 Mathematics cases and 7 Portuguese cases with G2 = 0 also have G3 = 0. This pattern helps distinguish persistent zero grades from other zero-grade patterns.

TYPE B

G3 = 0, G2 > 0

Investigates cases where a positive second-period grade is followed by a zero final grade. This category contains 25 Mathematics cases and 8 Portuguese cases, making it important to analyze separately from Type A.

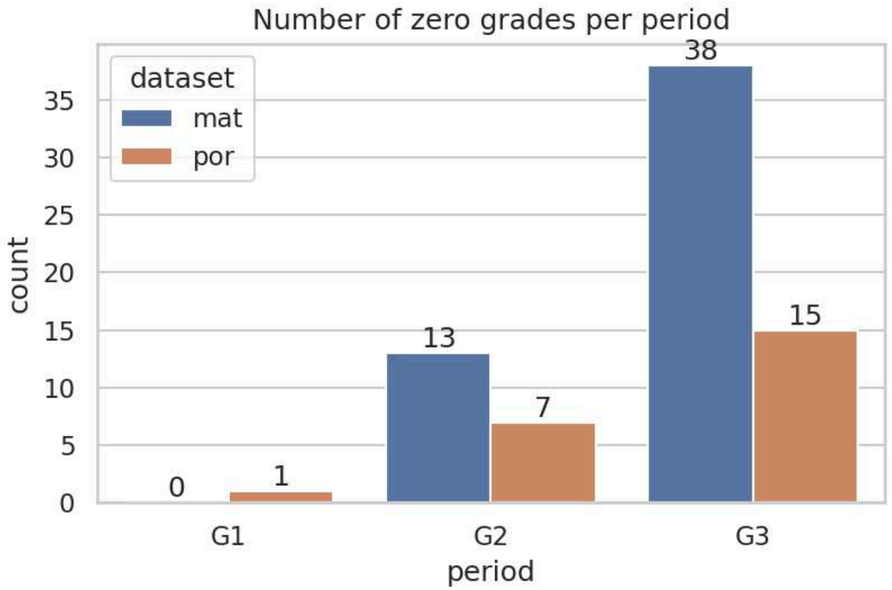
Anomaly

G1 =0, G2>0, G3 > 0

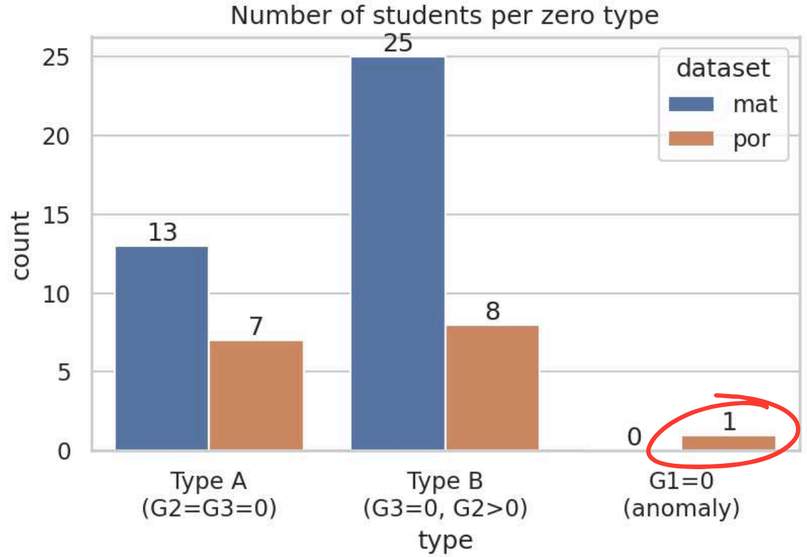
Examines an unusual grade sequence in the Portuguese dataset: one student has G1 = 0, G2 = 11, and G3 = 11. This case illustrates that a zero grade in the first period does not necessarily persist in subsequent assessments.

Discussion

Understand the pattern of zero grades



- Zeros are **almost absent** in G1
- Zeros increase from G2 to G3 (Math: 13 → 38, Portuguese: 7 → 15).
- Every G2 = 0 continues to G3 = 0
- Not every G3 = 0 starts at G2: 25 Math and 8 Portuguese students still have a valid G2.

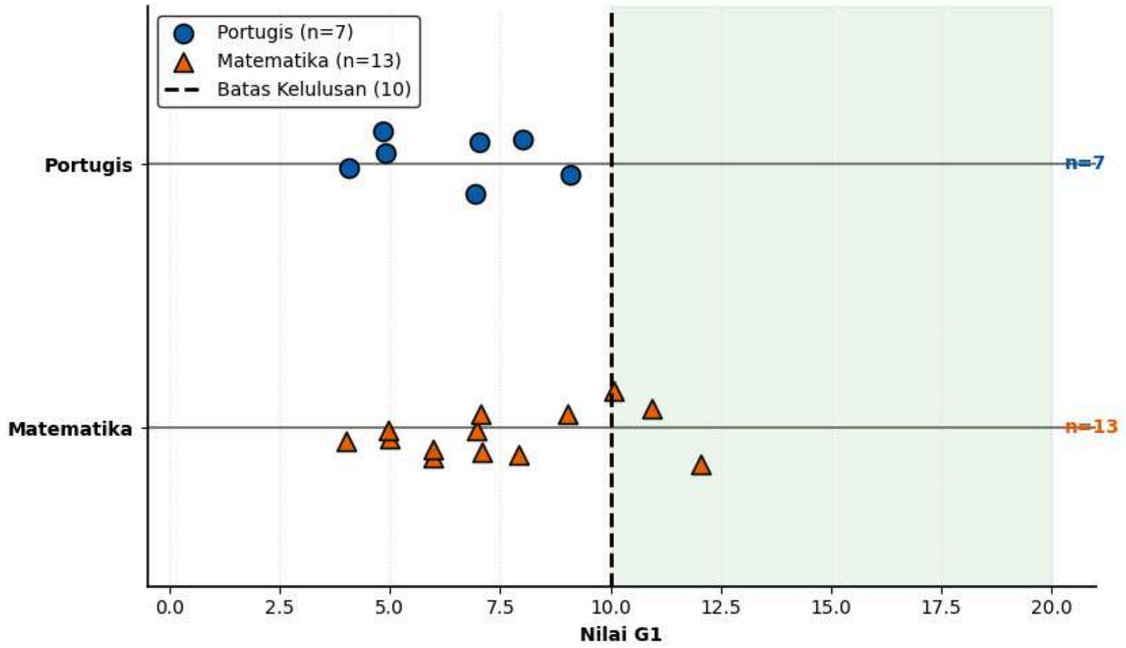


- Type A: G2 = 0 and G3 = 0 (Math 13, Portuguese 7).
- Type B: G3 = 0 with G2 > 0 (Math 25, Portuguese 8).
- Anomaly: one G1 = 0 case in Portuguese (G2 = G3 = 11).

Zeros follow a **one-directional pattern**: a zero in G2 always carries into G3, but a zero in G3 can appear without an earlier zero.

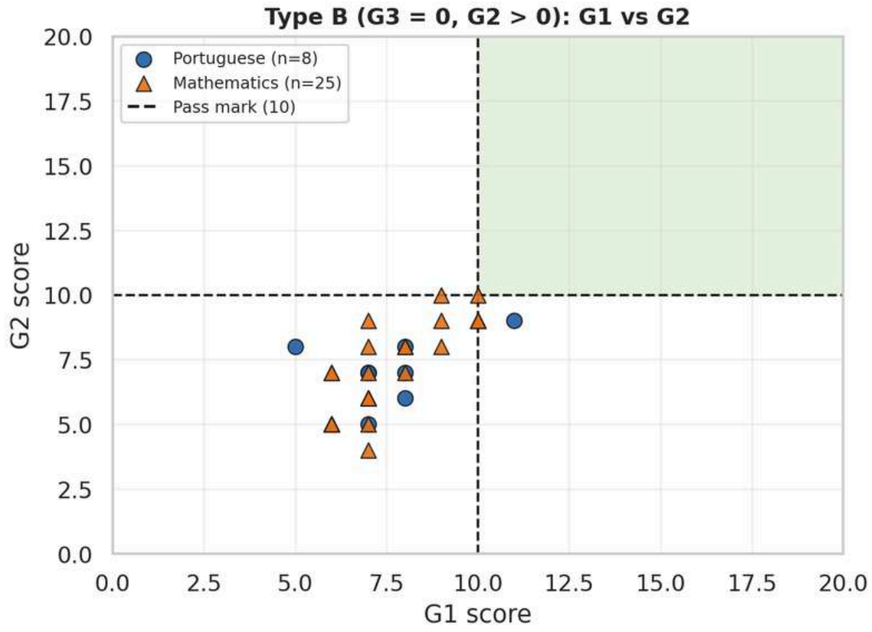
Discussion

Type A



- **Portuguese : All students had G1 scores below 10, ranging from 4 to 9.**
- **Mathematics : 10 out of 13 students had G1 scores below 10, with a maximum score of 12.**

Type B



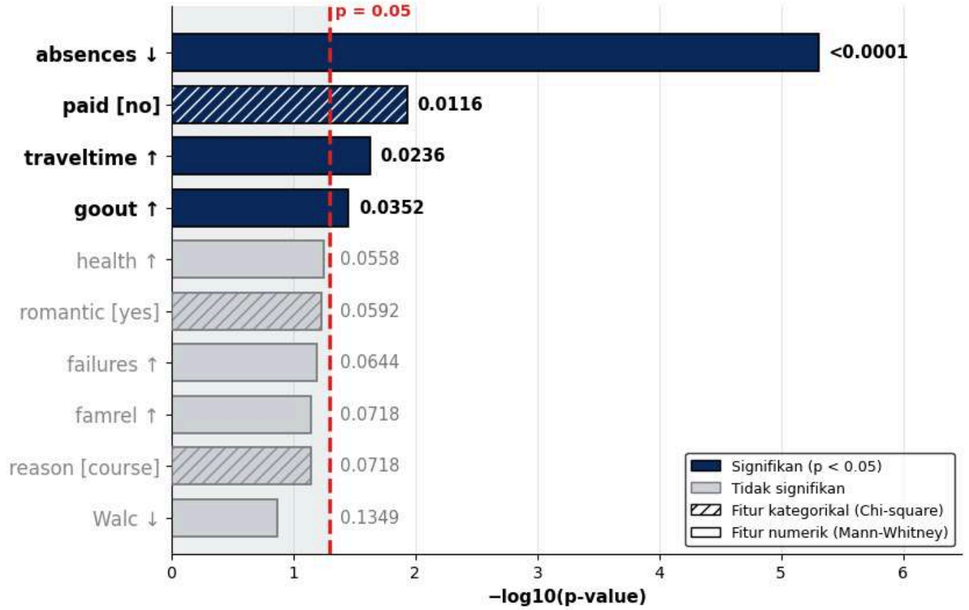
- **Portuguese : 7 of 8 students had G1 below 10, and the highest G2 score was only 9.**
- **Mathematics : 21 of 25 students had G1 below 10, and the highest G2 score was 10.**

Zeros **rarely appear out of nowhere**: most students were already below the pass mark before their grade became zero.

Features Associated with Zero Grades

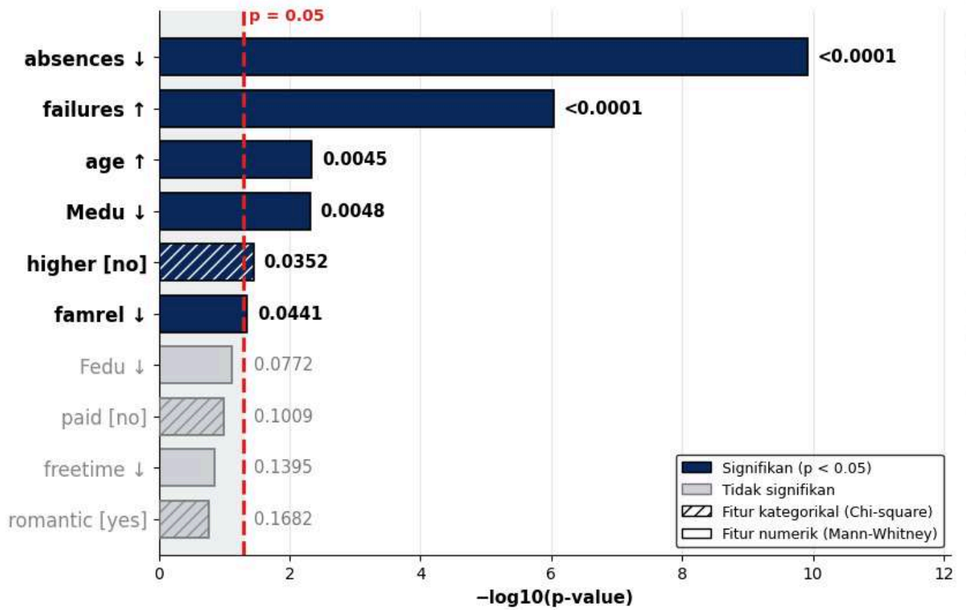
Mathematics

Type A



- **Absences** : Fewer absences
- **Paid classes** : Only 1 of 13 take paid classes vs 47% of others
- **Travel time** : Longer trip to school
- **Goout** : Slightly higher for zero-grade students Go out with friends more often

Type B

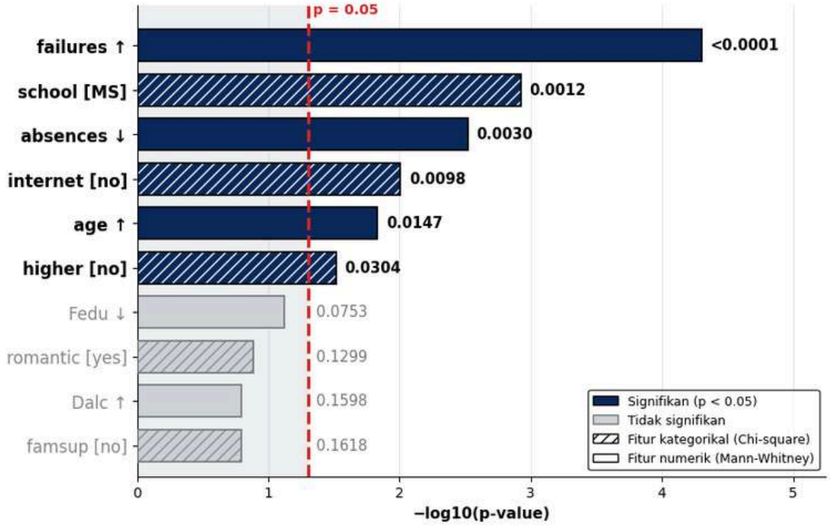


- **Absences** : Fewer absences
- **Failures** : More past failures
- **Age** : older (17.4 vs 16.7 years)
- **Medu** : Lower Maternal Education
- **Others** : 4 of 25 do not plan higher education (vs 4%); weaker family relationships (3.56 vs 3.97)

Features Associated with Zero Grades

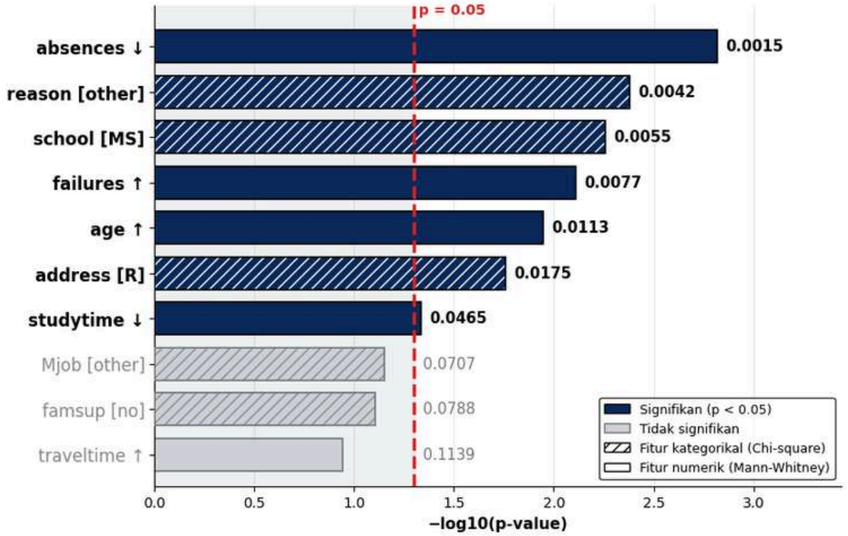
Portugis

type A



- **Failures** : More past failures
- **School** : 100% attend MS schools
- **Absences** : 0 absences
- **Others** : More likely to lack home internet, be older, and not plan higher education

Type B



- **Absence** : Fewer absences
- **Reason** : reason for choosing school differs
- **School** : which school MS dominates
- **Failures** : More past failures
- **Others** : Older, shorter study time, home address type

Absences is an important feature associated with zero grades. However, this relationship cannot be confirmed, as students who do not attend classes may also have no attendance records. Therefore, zero grades may not necessarily be caused by absenteeism.

Discussion

Sensitivity of Absence and Academic Grades to the Presence of 0 Grades

Scenario	Treatment	Students Removed	Purpose
S0	All students	None	Baseline
S1	Exclude Type A	Math = 13, Portuguese = 7	Sensitivity to zeros that begin at G2
S2	Exclude Type B	Math 25, Portuguese 8	Sensitivity to zeros that appear only at G3
S3	Exclude Type A & B	Math 38, Portuguese 15	Result when all zero-grade students are removed

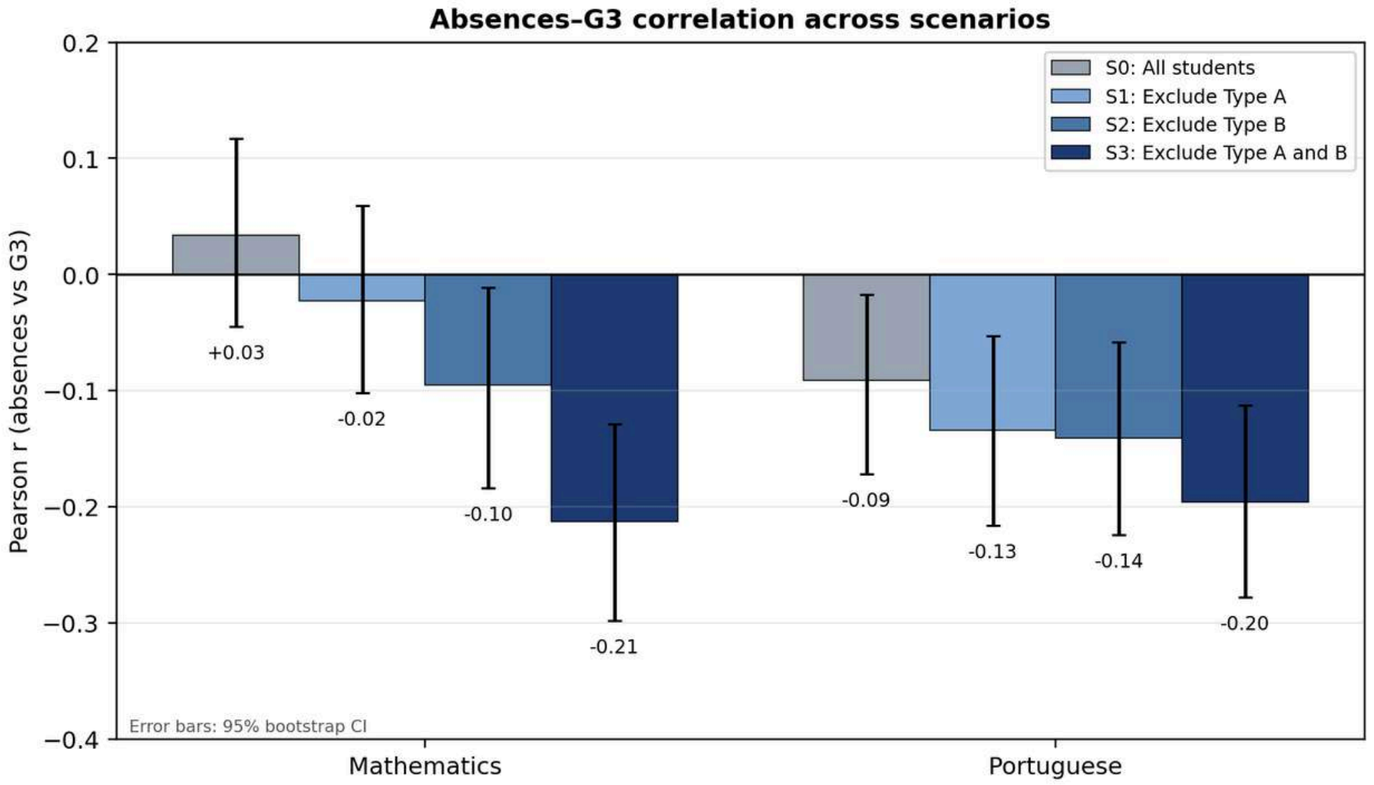
Method

- Correlation between absences and G3 computed for each scenarios
- Pearson r with 95% bootstrap CI
- Spearman ρ as a robustness check.

Type A and Type B are removed separately and together to see which zeros matter most.

Discussion

Sensitivity of Absence and Academic Grades to the Presence of 0 Grades



- **Mathematics:** r moves from +0.03 (S0, all students) to −0.21 (S3, all zeros excluded).
- **Portuguese:** r moves from −0.09 (S0) to −0.20 (S3).
- **Excluding Type A** only (S1) changes little in Mathematics (−0.02), while **excluding Type B** only (S2) goes further (−0.10).
- The negative association becomes clear only when both zero types are excluded.

Excluding students with zero final grades reveals a stronger **negative correlation** between absences and G3 in both subjects, particularly in Mathematics. This suggests that the observed correlation is sensitive to how zero final grades are treated.

Conclusion

1. **Distinct zero-grade patterns were identified**

Zero final grades do not always appear suddenly. Some students already had low grades in G1 and/or G2 before reaching G3 = 0, while others had positive earlier grades before their final grade dropped to zero. This suggests that zero final grades follow different patterns across students.

2. **Zero-grade students exhibit different characteristics**

Students with zero final grades tend to differ from other students in several observed characteristics, including previous failures, school-related factors, educational support, and study habits. These patterns vary between Mathematics and Portuguese.

3. **The absences-grade correlation is sensitive to zero-grade treatment**

After excluding students with G3 = 0, the correlation becomes more negative in Mathematics (r = -0.21) and Portuguese (r = -0.20). This indicates that the observed relationship between absences and final grades depends on how zero-grade observations are handled.

Examining zero-grade patterns is important for interpreting student performance data. However, the available dataset does not establish why these zero grades occur, so they cannot automatically be classified as recording errors or invalid grades.

REFERENCES

Cortez, P. (2008). Student Performance [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

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Discussion

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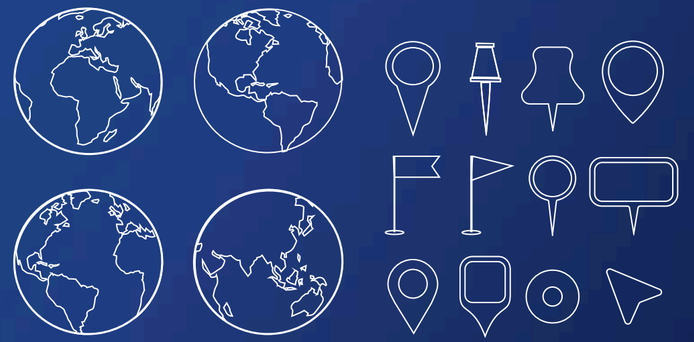
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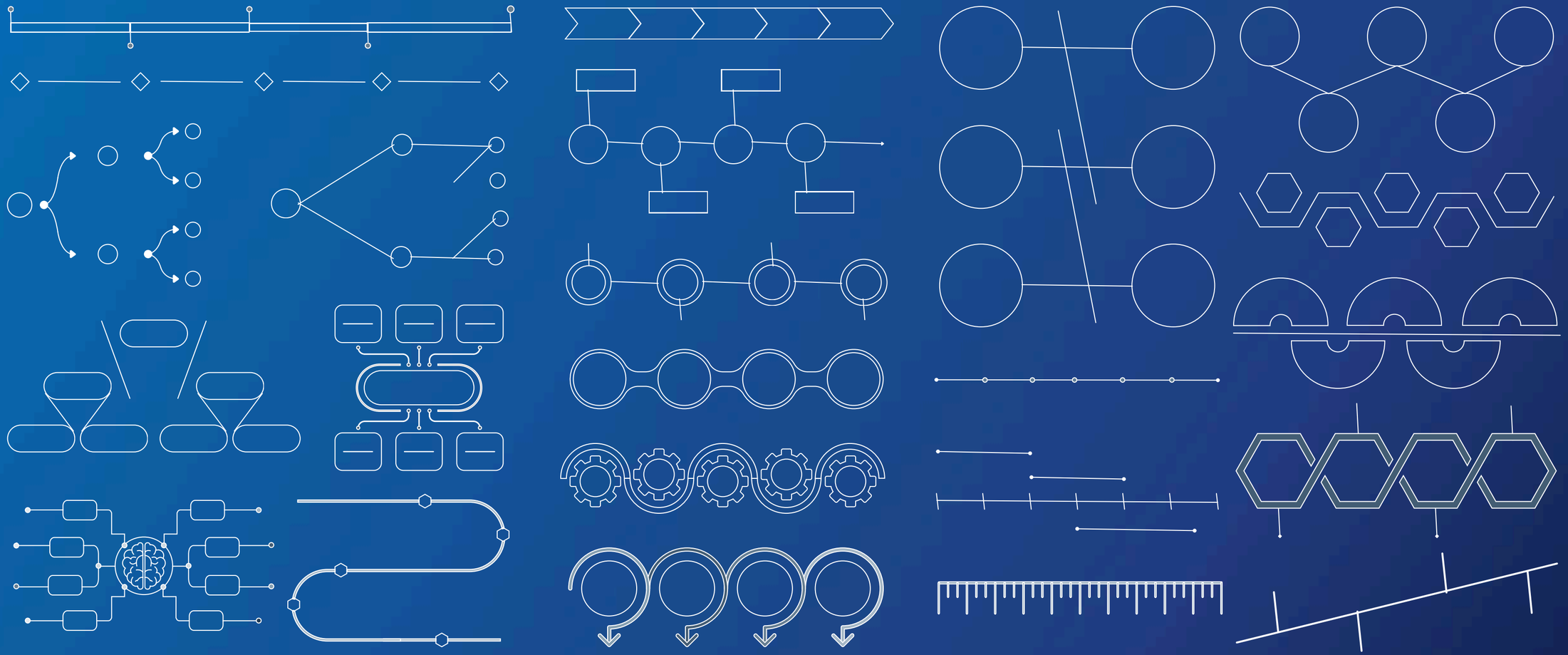
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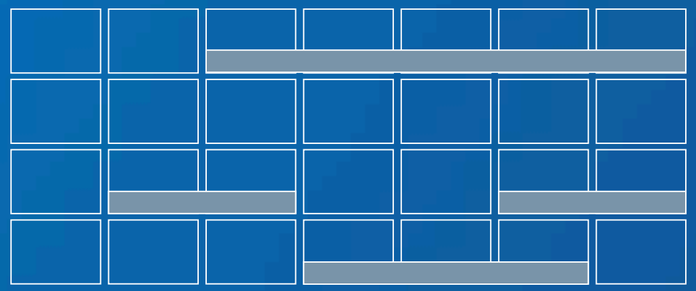


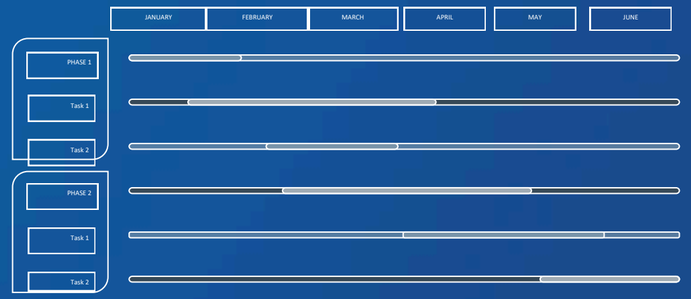



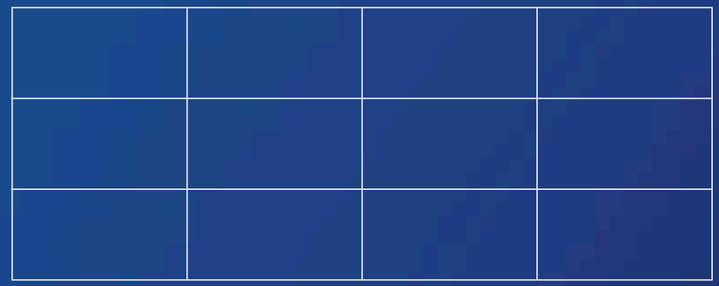
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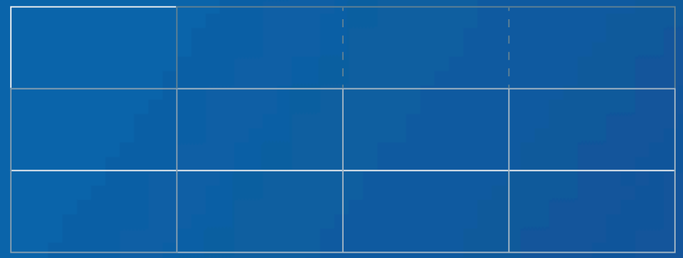


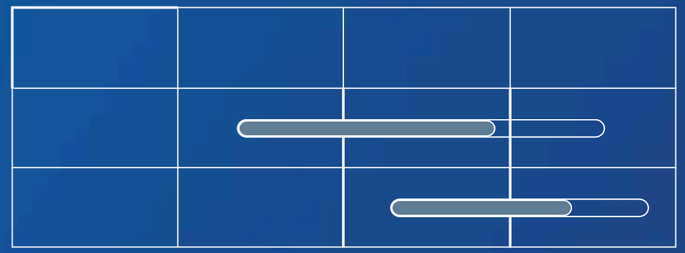
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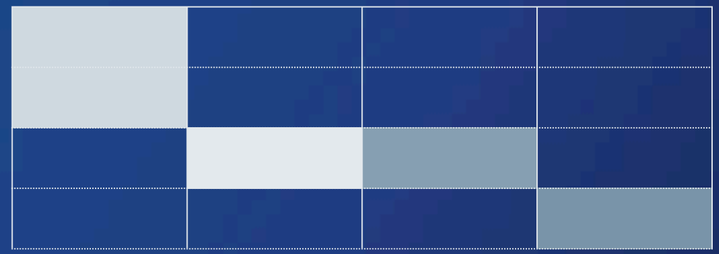


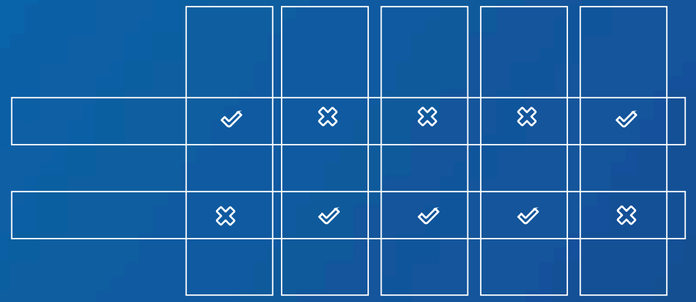


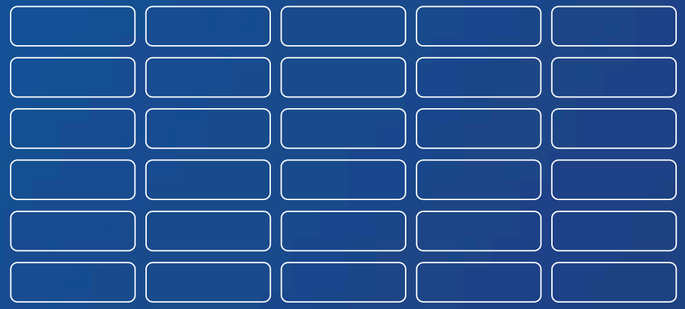


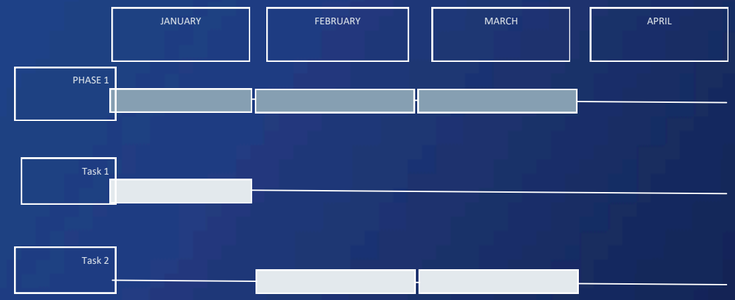




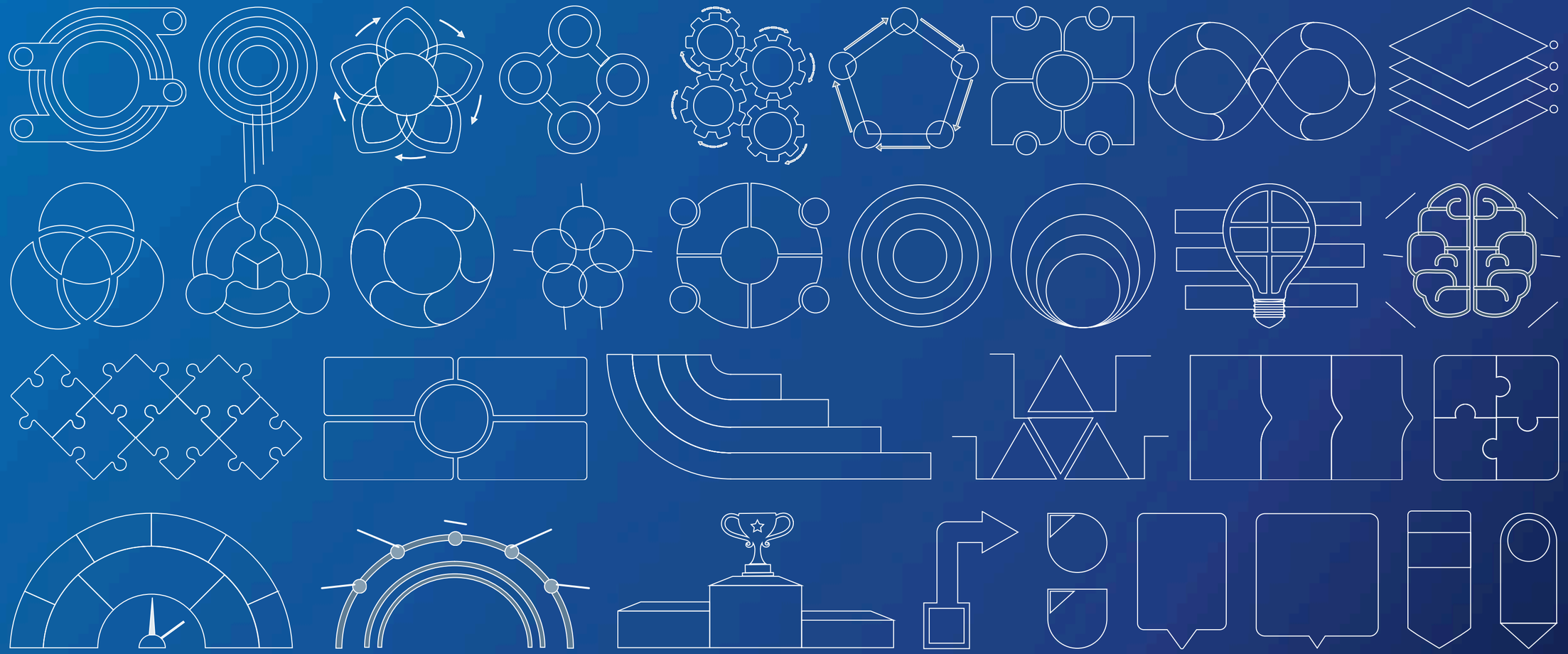


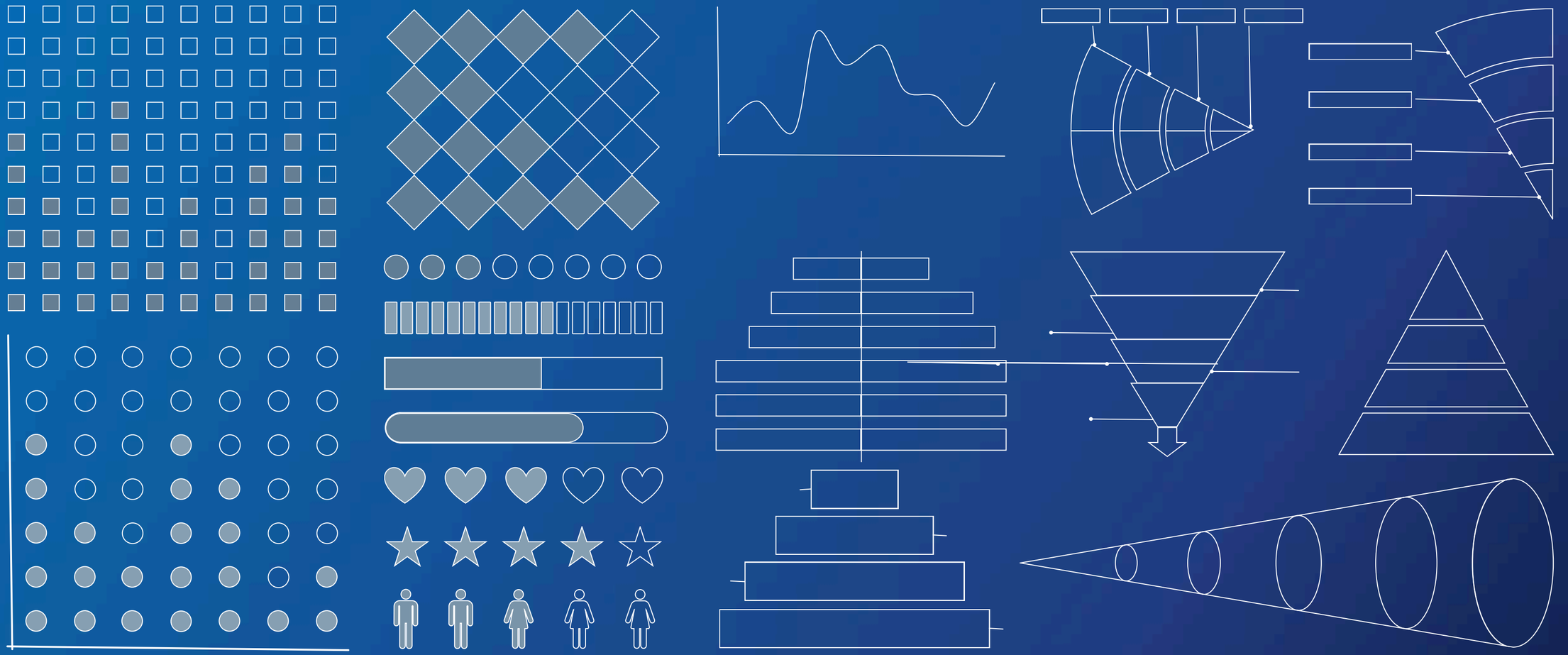






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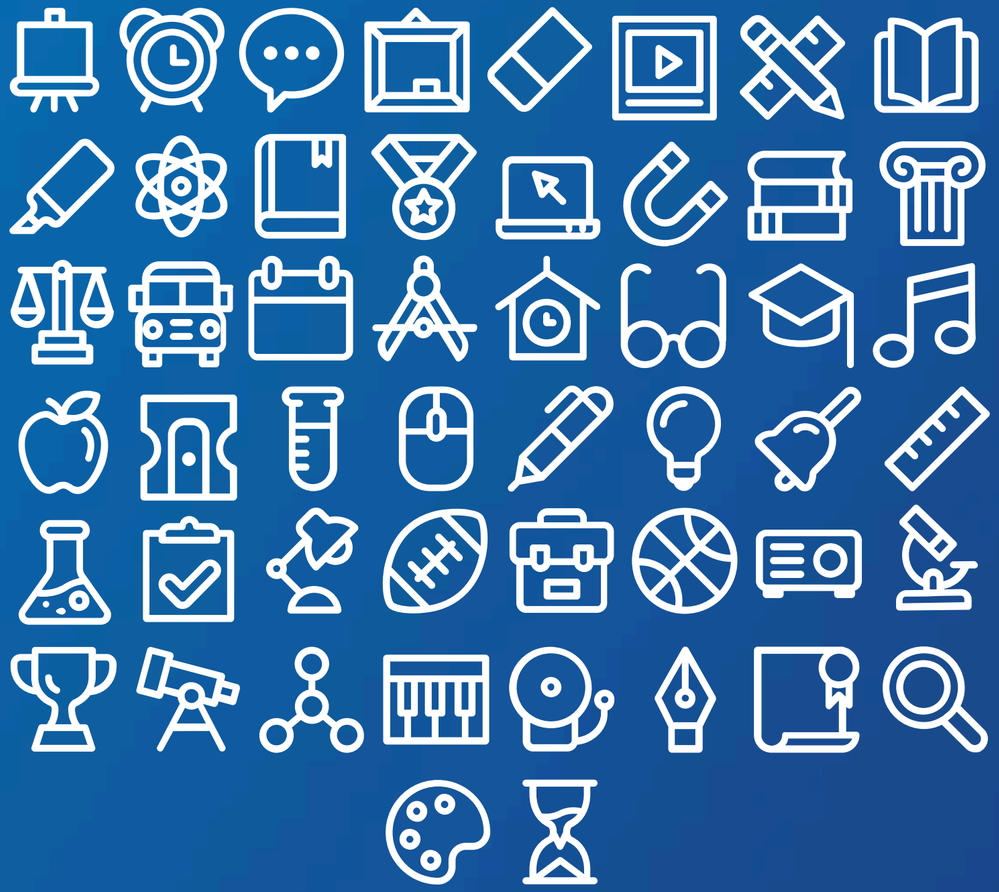
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